Lecture 1:
Key Concepts and Terms

Understand the idea. Work through an example.

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Start with the question, then the calculation

1. **Understand the idea.** What decision are we trying to make?
2. **Follow the example.** Use a calculator and do one step at a time.
3. **Explain the answer.** What does the number tell a manager?

The first part uses small numbers and everyday language. The calculation reference at the back introduces the symbols and fuller formulas.

Remember: This is a learning sequence, not a change to assessment requirements. Use the full lecture, practice questions and required reading too.

A percentage is an amount out of 100

The idea: A 10% return means earning £10 for every £100 invested.

1. Write 10% as a decimal: $10 \div 100 = 0.10$.
2. Find the gain: £100 $\times$ 0.10 = £10.
3. Add it to the starting amount: £100 + £10 = £110.

What this means: Multiplying by 1.10 keeps the original money *and* adds 10%. Multiplying by 0.10 finds only the gain.

A timeline tells us when money moves

The idea: A **cash flow** is money paid or received. Its date matters.

	Today	In one year
What happens	Pay £100	Receive £110
On a timeline	−100	+110
Date label	0	1

What this means: A minus sign means money leaves you. A plus sign means money comes to you. Date 0 simply means today.

Remember: Compare amounts at the same date. £110 next year is not automatically worth £110 today.

Compounding means earning interest on interest

The idea: With **compound interest**, each year's interest joins the money earning interest next year.

Example: £100 invested at 5% a year

1. End of year 1: $100 \times 1.05 = £105$.
2. End of year 2: $105 \times 1.05 = £110.25$.

What this means: The extra 25p is interest earned on the first year's £5 interest.

Remember: **Simple interest** pays interest only on the original £100: £5 each year, giving £110 after two years.

Formula and notation reference →

Discounting works backwards from future money

The idea: Present value (PV) asks: "How much would I need today to produce that future payment?"

Example: £121 due in two years; a 10% annual rate

1. Move back one year: $121 \div 1.10 = £110$.
2. Move back one more year: $110 \div 1.10 = £100$.

What this means: £100 today can grow into £121 in two years. They have the same value at this rate.

Remember: Moving forward: multiply. Moving backward: divide. **Future value (FV)** is the amount at the future date.

Formula and notation reference →

Several payments? Value each one, then add

The idea: Payments arriving on different dates need separate calculations.

Example: use a 10% annual discount rate

Payment	Calculation	Value today
£110 in one year	$110 \div 1.10$	£100
£121 in two years	$121 \div 1.10 \div 1.10$	£100

What this means: The payments together are worth £200 today. The later payment needs two steps backwards.

Remember: Use a rate with the same period as the payments: annual with years, monthly with months.

Formula and notation reference →

An annual quote is not always the annual return

The idea: The **effective annual rate** includes interest earned on interest within the year.

Example: a 12% annual quote, compounded twice a year

1. The rate for each half-year is $12\% \div 2 = 6\%$.
2. After six months: $100 \times 1.06 = £106$.
3. After one year: $106 \times 1.06 = £112.36$.

What this means: The effective annual return is 12.36%, not 12%.

Remember: UK **APR** concerns borrowing costs, including relevant charges; savings **AER** concerns the annual equivalent return. Check the quotation convention.

Formula and notation reference →

Inflation changes what your money can buy

The idea: **Nominal** means money amounts. **Real** means purchasing power: what the money buys.

Example: your savings earn 5%; prices rise 5%

1. Your £100 becomes £105.
2. A basket that cost £100 now costs £105.
3. You can still buy one basket: your real gain is zero.

What this means: More pounds do not always mean more purchasing power.

Remember: Use nominal cash flows with a nominal rate, or real cash flows with a real rate. Keep the two consistent.

Formula and notation reference →

An annuity is a fixed series of equal payments

The idea: An **ordinary annuity** pays the same amount at the end of each period, for a fixed number of periods.

Example: £100 in one year and £100 in two years; rate 10%

1. First payment today: $100 \div 1.10 = £90.91$.
2. Second payment today: $100 \div 1.10 \div 1.10 = £82.64$.
3. Add them: $90.91 + 82.64 = £173.55$.

What this means: The annuity formula is a shortcut for these separate calculations. You already know the underlying method.

Formula and notation reference →

A perpetuity is a payment that continues forever

The idea: A **level perpetuity** pays the same amount every period, with no end date. It is a valuation model.

Example: £100 a year forever; first payment next year; rate 5%

1. A £2,000 fund earns £100 a year at 5%.
2. Paying out that interest leaves the £2,000 fund intact.
3. So the payment stream is worth £2,000 today.

What this means: The shortcut is "annual payment divided by the annual rate": $100 \div 0.05 = 2,000$. The rate must be positive.

Formula and notation reference →

Earlier payments are worth more at a positive rate

The idea: An **annuity due** begins now. A **delayed stream** begins later. Always locate the first payment.

Example: two £100 payments; rate 10%

Payment dates	Calculation	Value today
Years 1 and 2	From previous example	£173.55
Today and year 1	$100 + 100 \div 1.10$	£190.91

What this means: Same total received, different timing, different value.

Remember: An annuity or perpetuity formula first gives a value *one period before* the first payment. Discount further if needed.

Formula and notation reference →

NPV asks how much value a project adds

The idea: Net present value (NPV) is the value today of money received, less the value today of money paid.

Example: pay £100 today; receive £121 next year; rate 10%

1. Value the receipt today: $121 \div 1.10 = £110$.
2. Subtract the cost today: $110 - 100 = £10$.

What this means: The project adds £10 of value today after allowing for the required return.

Remember: Include **incremental cash flows**: payments that change because you undertake the project, including later costs.

Formula and notation reference →

The required return is what investors give up elsewhere

The idea: The **opportunity cost of capital** is the return available on an alternative with comparable risk. It sets the required return.

Example: a comparable investment offers 10%

1. Investing £100 must offer £110 after one year just to match it.
2. A project offering £110 has NPV zero.
3. One offering £121 has NPV £10 at that same rate.

What this means: An **economic rent** is a return above opportunity cost. Ask what advantage could sustain it against competition.

Remember: Match the discount rate to the project's risk, currency and inflation assumptions.

The NPV decision depends on what you can choose

The idea: A positive NPV means estimated value is created. The available choices and constraints still matter.

Project	NPV
A	£20
B	£12

- **Independent:** you can do both. Accept both if feasible and unconstrained.
- **Mutually exclusive:** you can choose only one. Choose A.
- If both NPVs are negative, doing neither gives the better result.

IRR expresses a project's return as a percentage

The idea: The **internal rate of return (IRR)** is the discount rate at which the project just breaks even: NPV equals zero.

Example: pay £100 today; receive £121 next year

1. The gain is $121 - 100 = £21$.
2. Relative to the £100 invested: $21 \div 100 = 21\%$.
3. At a 21% discount rate, £121 is worth exactly £100 today.

What this means: For a standard investment with an initial cost followed by receipts, IRR above the required return means positive NPV.

Remember: The one-year calculation is simple. Several years require finding the rate that makes the full NPV zero.

A higher percentage need not create more value

The idea: NPV measures value in pounds. IRR measures a percentage. They answer different questions.

Example: two one-year projects, valued at a 10% required return

	Cost	Receipt	IRR	NPV
A	£100	£130	30%	£18.18
B	£1,000	£1,200	20%	£90.91

What this means: If you can choose only one, have enough funding and the risks are comparable, B creates more value despite its lower IRR.

Remember: Different payment dates can also change rankings. Later costs can produce more than one IRR. Use NPV at the appropriate rate.

Payback asks how soon the outlay is recovered

The idea: The **payback period** is the time it takes receipts to cover the starting cost. Compare it with a chosen cutoff.

Example: pay £100; receive £60 at each year-end

1. After year 1, you have received £60: not enough.
2. After year 2, you have received £120: cost recovered.

What this means: With year-end receipts, payback occurs at the end of year 2.

Remember: **Discounted payback** values receipts today first. At 10%, the two receipts are worth £104.13 in total. Both methods ignore payments after recovery.

Book value is an accounting number

The idea: Depreciation spreads an asset's cost over its useful life. **Book value (BV)** is the recorded value left after depreciation.

Example: an asset costs £500 and is depreciated evenly to zero over five years

1. Annual depreciation: $500 \div 5 = £100$.
2. Book value after one year: $500 - 100 = £400$.

What this means: The £100 depreciation is not a new cash payment. The purchase payment has already happened.

Remember: Book value need not equal the price someone would pay for the asset today.

AAR uses accounting profit, not cash flow

The idea: Average accounting return (AAR) compares average annual net income with average book value. Net income is profit after depreciation and tax.

Example: average annual net income £50; average BV £250

1. Divide profit by average BV: $50 \div 250 = 0.20$.
2. Express as a percentage: $0.20 \times 100 = 20\%$.

What this means: If the stated AAR target is 15%, this project passes that accounting test.

Remember: This does not prove positive NPV. AAR ignores payment dates and depends on accounting choices. State the denominator consistently.

PI expresses value received per pound invested

The idea: The **profitability index (PI)** compares the value today of future receipts with the initial cost.

Example: spend £100 for receipts worth £110 today

1. NPV is $110 - 100 = £10$.
2. PI is $110 \div 100 = 1.10$: £1.10 of value per £1 spent.

What this means: PI above 1 and NPV above zero give the same accept/reject answer here.

Remember: Some sources use a **net index**: $10 \div 100 = 0.10$, with a threshold of zero. Ratios can rank projects differently from total value.

A positive forecast still needs to be challenged

The idea: A calculation is useful only if its assumptions are credible.

- **Sensitivity analysis:** change one assumption. If next year's receipt falls from £121 to £110, our £100 project has NPV zero at 10%.
- **Scenario analysis:** change related assumptions together, such as lower sales and higher costs.
- **Feasibility:** check that the business has the funding and resources to deliver the project.

What this means: Managers need both a value estimate and a credible explanation of how the project will work.

Keep these questions attached to the terms

Term	The question it answers
PV	What is that future payment worth today?
FV	What will today's amount grow into?
NPV	How much value does the project add today?
IRR	At what discount rate does it break even?
Payback	How soon is the starting outlay recovered?
AAR	How much accounting profit per pound of average BV?
PI	How much present value per pound initially invested?

Check your understanding before using the shortcuts

1. Why is £110 next year worth £100 today at a 10% rate?
2. What does an NPV of £10 tell a manager?
3. Why might you choose a project with a lower IRR?
4. Why is depreciation different from a new cash payment?
5. Why can a quick payback still hide a poor investment?

Answer in words first. Then use one small example to support each answer.

Remember: Draw the dates, do one step at a time, and explain what the answer means.

Calculation reference: translate the symbols into words

The following pages support the calculations in the full lecture.

1. Identify the payment amounts and dates.
2. Read what each symbol stands for.
3. Substitute the numbers before reaching for a calculator.
4. Check the answer against the intuition from the first part.

Remember: You do not need to absorb all the formulas at once. Use the relevant page when working on that type of problem.

Reference: one amount moving through time

In words: multiply by the growth factor once for each period.

$$FV = PV(1+r)^n \qquad PV = \frac{FV}{(1+r)^n}$$

PV: value today; FV: future value; r: rate per period; n: number of periods.

Read the power: $(1.10)^2$ means 1.10×1.10.

Example: $100(1.10)^2 = £121$. Reverse it: $121/(1.10)^2 = £100$.

Remember: Rates can be negative. At -1%, £99 next year is worth £100 today. Simple interest instead uses $FV = PV(1+nr)$.

Reference: the large sigma means “add these up”

In words: value each receipt or payment today, then add.

$$NPV = CF_0 + \sum_{t=1}^{N} \frac{E[CF_t]}{(1+r)^t}$$

CF_0: cash flow now; CF_t: cash flow at date t; N: final date; $E[CF_t]$: expected cash flow; r: rate per period.

Example: pay £100 now; receive £60 at each of the next two year-ends; rate 10%.

$$NPV = -100 + \frac{60}{1.10} + \frac{60}{1.10 \times 1.10} = £4.13$$

Remember: The discount factor $(1+r)^{-t}$ turns a future payment into today’s value.

Reference: level and growing perpetuities

In words: a growing perpetuity pays forever, with each payment changing by the same percentage.

$$PV = \frac{C}{r} \quad \text{(level)} \qquad PV = \frac{C_1}{r - g} \quad \text{(growing)}$$

C: fixed payment; C_1: first payment, one period ahead; r: discount rate; g: payment growth rate.

Example: first payment £100; rate 5%; growth 2%. Payments start 100, 102, 104.04. Value: $100 \div (0.05 - 0.02) = £3,333.33$.

Remember: The level formula needs $r > 0$; the growing formula needs $r > g$ and $r, g > -1$. Negative growth is allowed.

Reference: the annuity shortcut

In words: these formulas combine equal payments at dates 1 to N.

$$PV = \frac{C}{r}\left[1 - (1+r)^{-N}\right] \qquad FV_N = \frac{C}{r}\left[(1+r)^N - 1\right]$$

C: payment; r: rate per period; N: number of payments. A negative power means divide: $(1+r)^{-N} = 1/(1+r)^N$.

Example: $C = 100$, $r = 0.10$, $N = 2$ gives PV £173.55 and value at date 2 of £210.

Remember: For an annuity due, multiply ordinary-annuity PV by $(1+r)$. At a zero rate, value is simply $N \times C$. A delayed stream must be discounted further from the formula's valuation date.

Reference: convert a quoted annual rate

In words: find the rate for one compounding period, then compound for a full year.

$$\text{Periodic rate} = j/m \qquad \text{Effective annual rate} = (1 + j/m)^m - 1$$

j: nominal annual quote; m: compounding periods per year.

Example: 12% quoted annually, compounded monthly.

1. Monthly rate: $0.12 \div 12 = 0.01$ (1%).
2. Annual growth factor: 1.01^{12}.
3. Subtract 1: $1.01^{12} - 1 = 0.1268$, about 12.68%.

Reference: remove inflation from the growth factor

In words: divide money growth by price growth. This is the **Fisher relation**.

$$1 + r_{\text{real}} = \frac{1 + r_{\text{nominal}}}{1 + \pi}$$

r_{real}: purchasing-power return; r_{nominal}: money return; π ("pi"): inflation rate.

Example: nominal return 3%, inflation 2%.

1. Divide growth factors: $1.03 \div 1.02 = 1.009804$.
2. Subtract 1: 0.009804, or about 0.98% real return.

Remember: "Nominal" here concerns inflation. A nominal annual *quote* on the previous page concerns compounding conventions.

Reference: estimate IRR between two trial rates

Interpolation means estimating between two known points.

Example: NPV is $+300$ at 10% and -100 at 10.5%.

1. The full NPV fall is $300 + 100 = 400$.
2. Zero is $300 \div 400 = 75\%$ of the way down.
3. Move 75% across the rate gap: $10\% + 0.75 \times 0.5\% = 10.375\%$.

$$\widehat{IRR} = r_L + \frac{NPV_L}{NPV_L - NPV_H}(r_H - r_L)$$

L: lower trial rate; H: higher trial rate; the hat means an estimate.

Remember: The rate gap is 0.5 percentage points. NPV is curved, so this straight-line estimate is approximate.

Reference: a crossover is a tie between projects

The idea: The **crossover rate** makes two projects' NPVs equal. An **NPV profile** plots NPV against the discount rate.

Example: both projects cost £90 today

- A pays £110 in one year; B pays £121 in two years.
- At 10%, each receipt is worth £100 today: both NPVs are £10.
- At 5%, B wins; at 15%, A wins.

What this means: Equal NPVs do not have to be zero. To find the tie, subtract the two projects' cash flows at each date and find the IRR of that difference.

Remember: Use a common rate only where appropriate for both projects' risk.

Reference: later costs can produce more than one IRR

The idea: A **non-conventional** cash-flow pattern can switch from costs to receipts and back to costs, as with a final cleanup bill.

Example: pay 100 now, receive 230 next year, pay 132 a year later

$$NPV = -100 + \frac{230}{1+r} - \frac{132}{(1+r)^2}$$

Both a 10% rate and a 20% rate make NPV zero. There are two IRRs.

At a required return of 15%, NPV is approximately $+0.19$.

What this means: Do not select whichever IRR looks attractive. Calculate NPV at the appropriate required return. Multiple sign changes allow, but do not guarantee, multiple IRRs.

Reference: MIRR separates the money in and money out

Modified internal rate of return (MIRR) makes financing and reinvestment assumptions explicit.

1. Value the outflows today: call the positive cost total PV_{-}.
2. Grow the inflows to the final date: call that total FV_{+}.
3. Find the compound rate connecting the two over N periods.

$$MIRR = \left(\frac{FV_{+}}{PV_{-}}\right)^{1/N} - 1$$

Example: cash flows $-100, +60, +60$; both rates 10%. Cost today is 100; final inflows are $60(1.10) + 60 = 126$. MIRR is $\sqrt{1.26} - 1 = 12.25\%$.

Remember: The subscripts label inflows and outflows. PV_{-} is entered as a positive cost amount, not a negative number.

Reference: modified returns depend on the method

The idea: An **adjusted-cash-flow IRR** and terminal-value **MIRR** reshape the payments differently.

- The lecture's adjusted-IRR method moves later costs backwards and combines cash flows until the adjusted stream has one sign change.
- Terminal-value MIRR values costs today and receipts at the final date.

Lecture example: King Coal, using 20% in both constructions, gives adjusted IRR 22.05% and five-year terminal MIRR 20.38%.

What this means: Both can be calculated correctly. Always state the rates, horizon and method; NPV still measures value created.

Reference: borrowing reverses the usual IRR intuition

The idea: If you receive money now and pay later, IRR describes a financing cost. A higher percentage is not automatically better.

Example: receive £100 now; repay £120 next year

- IRR is 20%: the one-year borrowing cost.
- At a 10% opportunity cost, NPV is $100 - 120 \div 1.10 = -£9.09$. Reject.

Remember: Some streams have **no IRR**. With only positive receipts and no outflows, NPV stays positive at every finite rate above -100%. There is no zero to find.

Reference: a limited budget may require combinations

The idea: Capital rationing means funds are limited. If projects cannot be split, the highest PI need not identify the best set.

Example: budget £10; independent projects

Project	Cost	NPV	PI
A	£6	£4.20	1.70
B	£5	£3.00	1.60
C	£5	£3.00	1.60

What this means: A leaves £4, too little for either other project. B and C together fit the budget and create £6: more than A's £4.20.

Remember: PI can help with divisible, independent projects and one initial budget. If later costs occur, state the denominator convention.